BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 206 (January 2023)

Energy Conversion Laboratory

Final Project Report

Section: B1 Group: 05

Construction Of An Isolated Step-Down Transformer Of Certain kVA Rating With Different Specifications

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Academic Honesty Statement:

IMPORTANT! Please carefully read and sign the Academic Honesty Statement, below. Type the student ID and name, and put your signature. *You will not receive credit for this project experiment unless this statement is signed in the presence of your lab instructor.*

"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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Table of Contents

1 Abstract 1

2 Introduction 1

3 Design 2

3.1 Problem Formulation (PO(b)) 2

3.1.1 Identification of Scope 2

3.1.2 Literature Review 2

3.1.3 Analysis 3

3.2 Design Method (PO(a)) 3

3.3 Circuit Diagram 5

3.4 Hardware Design 5

4 Implementation 6

4.1 Description 6

4.2 Experiment and Data Collection 7

4.3 Data Analysis 7

4.4 Results 9

4.5 Sustainability and Environmental Impact Evaluation (PO(g)) 9

5 Reflection on Individual and Team work (PO(i)) 10

6 Project Management and Cost Analysis (PO(k)) 10

6.1 Cost Analysis 10

6.2 Project Management 11

7 References 11

1 Abstract

The report portrays the analysis of an isolated 220/110V step-down transformer project, detailing its design, construction cost, VA rating, and efficiency. This project has been done with an objective to constructing an efficient transformer that would be able to reduce voltage level while ensuring electrical isolation between primary and secondary coils.

First of all, the construction requirements of an isolated transformer have been discussed in the report. The design parameters of transformer include the selection of core materials, winding configurations, and insulation materials, which are crucial for achieving the desired voltage transformation ratio and electrical isolation.

Next the assembly of the core and windings have been described. Necessary calculations have been done for the given parameters (**220/110V** and **90% tapping**), which gives a detailed idea about bobbin size, total wire, and wire diameter. Special attention has been given to ensuring the transformer's compliance with relevant standards and regulations.

Moreover, the report covers the rigorous testing procedures applied to the transformer, including open circuit and short circuit tests. The results of these tests evaluated the current, voltage, turns ratio, and efficiency of the transformer.

Furthermore, the report discusses the transformer's practical applications, emphasizing its importance in various electrical systems for voltage reduction and safety enhancement. Additionally, it highlights the economic and environmental benefits of using such transformers.

To conclude, a step-down transformer with 90% tapping has been designed, constructed, and tested in this project. Special attention has been taken so that it meets the required specifications and standards. The successful designing and implementation of this project may be highlighted as a learning outcome of the EEE 206 course.

2 Introduction

An isolated step-down transformer is a specialized type of electrical transformer designed to

decrease the voltage level of an electrical circuit while providing electrical isolation between the input and output. This means that it can take a higher voltage input and produce a lower voltage output, all while preventing direct electrical contact between the two sides.

Transformers work on the principle of Faraday's Law of induction, converting electrical energy of one level to another level. Thus, the significance of designing a step-down transformer is immense, when low voltage level is required. For example, it is used to reduce the high-voltage electricity generated at power plants to lower, safer voltage levels suitable for distribution to homes, businesses, and industries.

As an alternative, auto transformers could be used to adjust voltage levels, but they do not provide the same level of electrical isolation as traditional transformers.

3 Design

3.1 Problem Formulation (PO(b))

The central problem this project addresses the construction an isolated step-down transformer tailored to meet specific voltage reduction. This project seeks to:

- Designing a core-type transformer with 200 kVA rating
- Creating a transformer capable of precise voltage reduction- 220V/110V, while maintaining electrical isolation between primary and secondary windings
- Maintaining 90% tapping at the secondary windings
- Evaluating the primary and secondary currents and voltages
- Ensuring maximum safety and high efficiency, and trying to complete the project within moderate cost

Failure to address this problem could result in compromised electrical safety, inefficient energy distribution, and reduced system performance.

3.1.1 Identification of Scope

Our project aims to create a transformer that can efficiently reduce voltage levels while providing electrical isolation between primary and secondary circuits. Its scope encompasses the design, construction, testing, and evaluation of an isolated step-down transformer.

3.1.2 Literature Review

Literature review reveals the importance of transformers in electrical systems,

emphasizes their role in voltage transformation and isolation. Relevant standards, such as those set by organizations like IEEE, highlight the necessity of compliance in transformer design and operation.

3.1.3 Analysis

The problem analysis takes into account the technical and economic dimensions. The technical aspect involves designing the transformer with specific voltage reduction 220V/110V and precise isolation characteristics. Safety considerations are vital to prevent electrical accidents and ensure compliance with safety standards. Economically solving this problem can lead to cost savings and improved efficiency.

3.2 Design Method (PO(a))

Knowledge of the theory course we studied was solely applied to solve the problem of this project.

The working method of our project is described below in short:

- First of all, bobbin was required. Searching in the market, we found a bobbin slightly bigger than our calculated size. We used that one.
- Next, we bought the E and I cores matching the size of bobbin.
- We also bought 'SWG27' copper wires.
- The High Tension (HT) side was first designed with 700 turns of these copper wires.
- After that, we used insulation tapes to separate the primary and secondary windings from each other.
- Then the LT side was designed with 350 turns; however, at 315 turn, an output wire was taken out to ensure 90% tapping.
- After completing all these wiring, insulation mask was again used.
- Cores were inserted into the bobbin.
- The exposed wires were kept safe with insulation tapes.

Mathematical knowledge of transformer was used to perform the following calculations:
Rating: 220/110V step down transformer
Apparent power: 200VA

Current rating:

Primary:	**Secondary:**
V_p =220V $I_{p,max} = \frac{200}{220}$=0.909A	V_s =110V $I_{s,max} = \frac{200}{110}$=1.81A

SWG of Wire Selection:

	SWG Number	**Current Capacity,A**	
		750 kcmil/A	**500kcmil/A**
Primary and secondary	SWG 24	0.645	0.968

Finding Bobbin Size:

Bobbin Area=0.16*$\sqrt{VA}$=0.16*$\sqrt{200}$=2.27 inch^2

Finding turns Ratio:

Turns per Volt, TPV=$\frac{1}{4.44fAB}$

Where,

Frequency, f=50Hz

Magnetic Flux density, B=0.97Wb/m^2

(Value of B from the standard and B-H curve of the core material)

Bobbin Area, A=2.27 inch^2=1.4645 x10^{-3} m^2

TPV = $\frac{1}{4.44fAB}$ =$\frac{1}{4.44*50*1.0345*1.4645*10^{-3}}$ = 3.17

Primary Winding Turns =220*3.17=697.4≈ 700

Secondary Winding Turns =110*3.17=348.7≈ 350

Total Wire Required:

Bobbin parameter =2*(L+W)

=2*(1.375+1.9375)

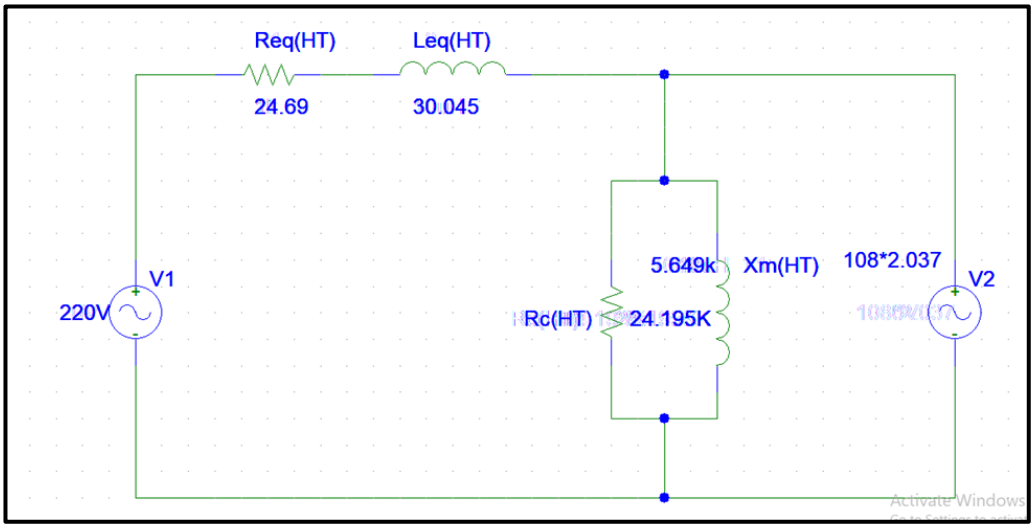
=6.625 inch

=0.1683m

Total wire of Primary = 0.1683*700= 117.81m

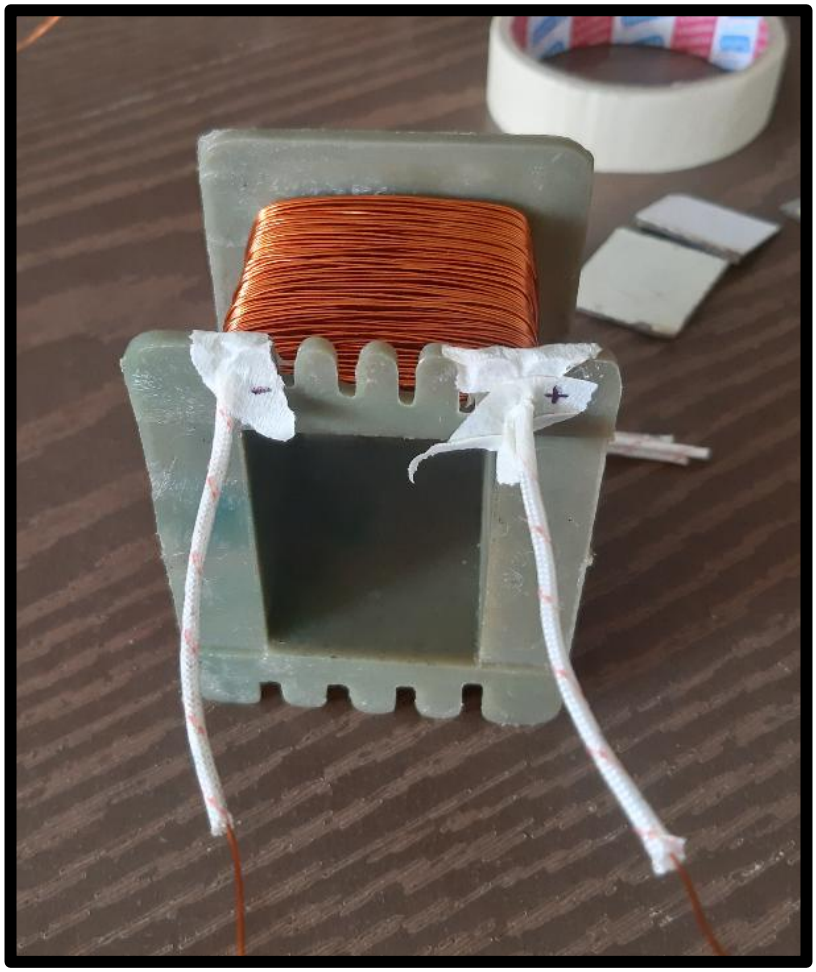
Total wire of secondary = 0.1683*350= 58.905m

3.3 Circuit Diagram



3.4 Hardware Design

Wiring of transformer:



Final Transformer**:**



4 Implementation

4.1 Description

From the calculations in section 3.2, we reached the following decisions about designing:

Rating: 220/110V step down transformer
Apparent power: 200VA
Bobbin Area=0.16*$\sqrt{VA}$=0.16*$\sqrt{200}$=2.27 inch^2

Primary Winding Turns =220*3.17=697.4≈ 700
Secondary Winding Turns =110*3.17=348.7≈ 350

Total wire of Primary = 0.1683*700= 117.81m
Total wire of secondary = 0.1683*350= 58.905m

4.2 Experiment and Data Collection

Turns ratio testing:

After applying rated voltage of 220V at primary and secondary,

We get,

V_p =220V

V_s =108V

Turns ratio, a = $\frac{220}{108}$ = 2.037

Open Circuit Test and Short Circuit Test:

Open circuit test was done at HT side of the transformer. We get the following data:

P_{oc}	2W
V_{oc}	220V
I_{oc}	0.04A

Short circuit test was done at HT side of the transformer. We get the following data:

P_{sc}	20W
V_{sc}	35V
I_{sc}	0.9A

Tapping:

Vp=220V

$V_{p(tapping)}$=198V

$\frac{Vp(tapping)}{Vp}$ x100%=$\frac{198}{220}$ x100% = 90%

This tapping is 90 % taps on its primary winding.

4.3 Data Analysis

From **open circuit test**,

Power factor angle, Θ =$\cos^{-1}\frac{Poc}{Voc*Ioc}$= $\cos^{-1}\frac{2}{220X.04}$ = 76.86°

Admittance, Y_E=$\frac{Ioc}{Voc}$ ∠-θ

=$\frac{0.04}{220}$∠-76.86°

=4.133x10^{-5} -1.77x10^{-4}j

=$\frac{1}{Rc}$ − $\frac{j}{Xm}$

$\frac{1}{Rc} = 4.13x10^{-5} \Rightarrow R_c = 24.195K\Omega$

$\frac{1}{Xm} = 1.77x10^{-4} \Rightarrow X_m = 5.649K\Omega$

From **short circuit test**,

Power factor angle, e, $\theta = \cos^{-1}\frac{Psc}{Vsc \times Isc} = \cos^{-1}\frac{20}{35*0.9} = 50.586^\circ$

Impedance, $ZSE = \frac{Vsc}{Isc}\angle\theta$

$= \frac{35}{0.9}\angle 50.586^\circ$

$= 24.69 + 30.045j$

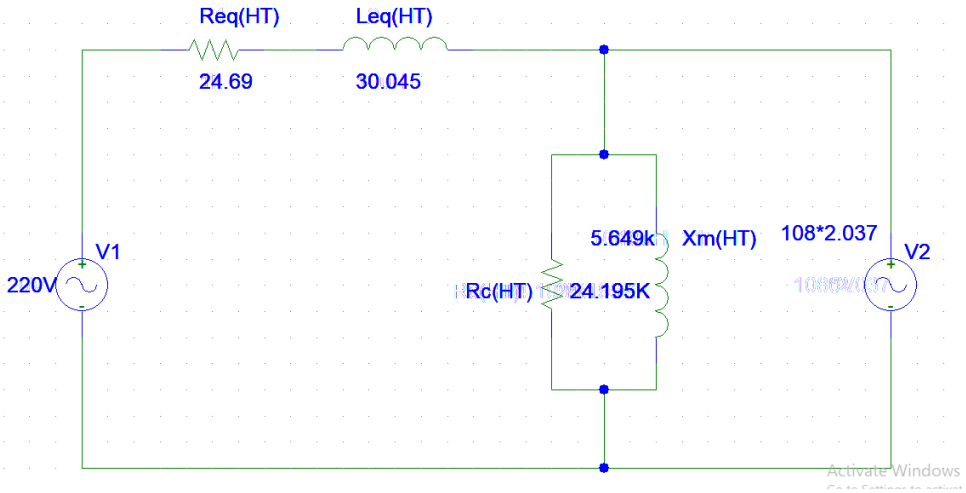
$= Req + j\,Xeq$

$Req = 24.69\Omega$

$Xeq = 30.045\ \Omega$

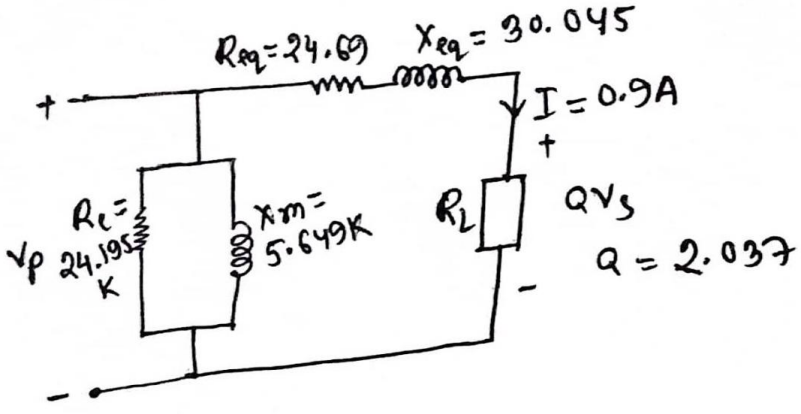
Note: Both the OCT and SCT are done with respect to HT side.

So, the final equivalent circuit of the transformer in terms of HT side/Primary side:



Theoretical calculation of Voltage Regulation and Efficiency:

From the equivalent circuit we can calculate voltage regulation and efficiency theoretically. In this case, we will find VR and efficiency for full load at unity power factor. Let, the transformer is supplying rated power(200W) a fully resistive load. Then from the HT equivalent circuit,



For fully load unity power factor current, I=0.9A.

Voltage Regulation calculation:

$V_p = aV_s + R_{eq}I + jX_{eq}I$
$= 220\angle 0^\circ + (24.69 + 30.045j) * (0.9\angle 0^\circ)$
$= 243.726\angle 6.37^\circ$

Voltage regulation, $VR = \frac{Vs,nl - Vs,fl}{Vs,fl} x100\%$

$= \frac{Vp - aVs}{aVs} x100\%$

$= \frac{24.726 - 220}{220} x100\%$

$= 10.78\%$

Efficiency calculation :

$P_{out} = VI = 200W$
Resistance,
$P_{cu,loss} = I^2R_{eq} = 0.909^2 x 24.69 = 20.4W$
$P_{core,loss} = (V^2/R_c) = ((243.726)^2)/(24.195x10^3) = 2.45W$

Efficiency, $\eta = \frac{Pout}{Pin} x100\%$

$= \frac{Pout}{Pout + Pcu,loss + Pcore,loss} x100\%$

$= \frac{200}{200 + 20.4 + 2.45} x100\%$

$= 89.746\% \approx 90\%$

So, the transformer has approximately 90% efficiency full load.

4.4 Results

V_p = 220 V; **V_s** = 108.5 V; **$V_{p(tapping)}$** = 197.9 V
Voltage Regulation = 10.78%
Efficiency = 89.746%

4.5 Sustainability and Environmental Impact Evaluation (PO(g))

*This section is mandatory to write if the course outcomes address PO(g) Read the PO Statement in the website first, before writing this section (*Program Outcomes and Program Educational Objectives | Department of EEE, BUET*)*

5 Reflection on Individual and Team work

Individual Contribution of Each Member

Every member of the team contributed greatly to the project. Initially, two teams were formed, one for theoretical calculation, and another visited Nawabpur and Dholaikhal to buy the materials according to the calculation.

After purchasing everything, we wired the bobbin and adjusted the core manually. Since the work was tiresome, we did it simultaneously. Some pictures of our work are attached to the report.

6 Project Management and Cost Analysis (PO(k))

6.1 Cost Analysis

This table contains all the expenses of the project:

Name of the Equipment	Unit	Cost(BDT)
Winding Wire	SWG 24:700gm	980
Transformer's core(E type & I type)	4kg	600
Bobin	1Unit	120
Mounting Hardward (Screw,Brackets etc.)	1pack of 4 units	50
Insulating Material (Tape,Tube)	1 unit each	60
Safety Equipment	-	50
transportation	-	200
Total	-	2060

6.2 Project Management

There were total six members in our group. We divided the total cost among ourselves. We first calculated required amount of wires, quality of products, etc, thus ensured to complete the project within minimum costing.

7 References

Electric Machinery Fundamentals, by Stephen J. Chapman



